

Lecture 1

Wireless Introduction

Transmitting voice and data using
electromagnetic waves in open
space

Electromagnetic waves

Travel at speed of light ($c = 3 \times 10^8$ m/s)

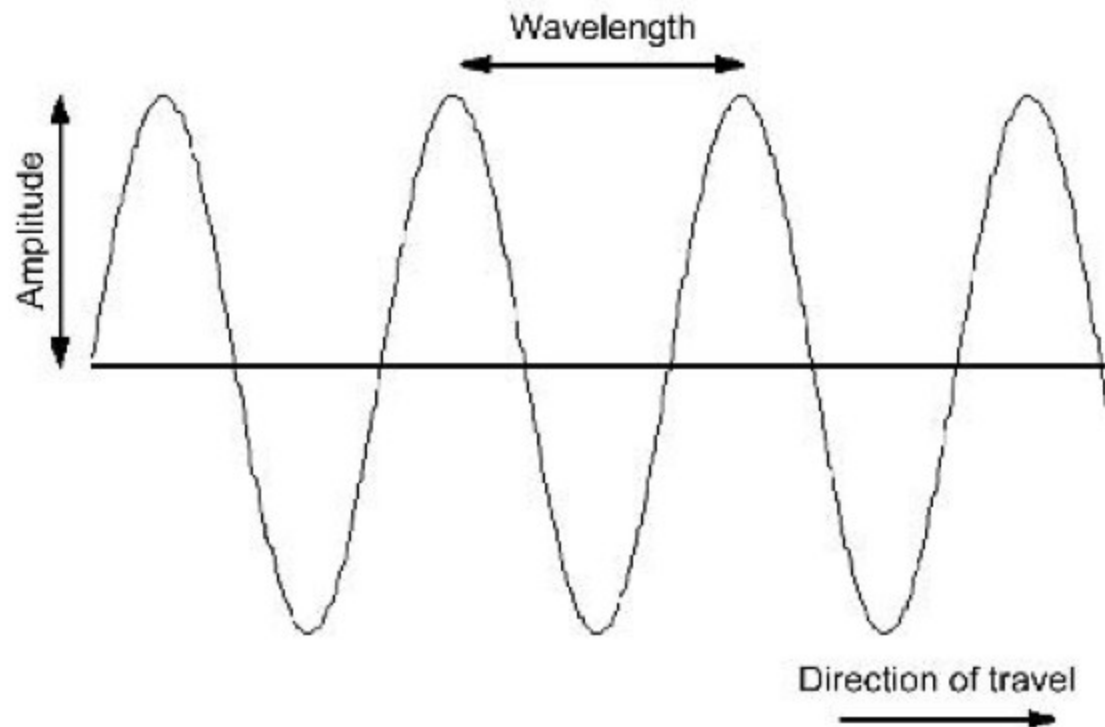
Has a frequency (f) and wavelength (λ)

$$c = f \times \lambda$$

Higher frequency means higher energy photons

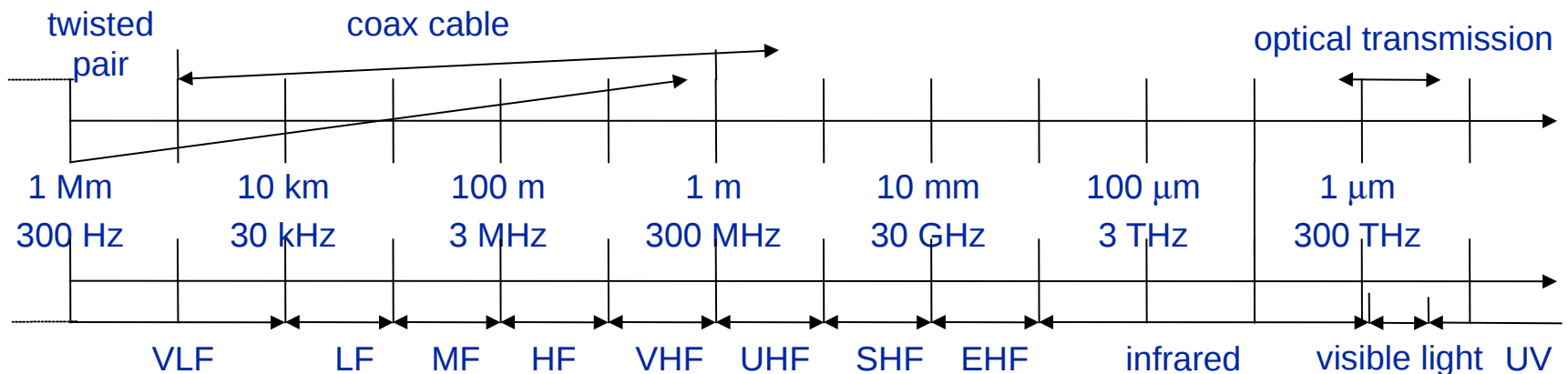
The higher the energy photon the more
penetrating is the radiation

Electromagnetic Spectrum



Frequencies for communication

- VLF = Very Low Frequency
 - LF = Low Frequency
 - MF = Medium Frequency
 - HF = High Frequency
 - VHF = Very High Frequency
 - UHF = Ultra High Frequency
 - SHF = Super High Frequency
 - EHF = Extra High Frequency
 - UV = Ultraviolet Light
- Frequency and wave length
 - $\lambda = c/f$
 - wave length λ , speed of light $c \cong 3 \times 10^8 \text{ m/s}$, frequency f



Various Radio bands and their uses

Frequency	Band name	Applications
< 3 KHz	Extremely Low Frequency (ELF)	Submarine communications
3 KHz -30 KHz	Very Low Frequency (VLF)	Marine communications
20 KHz -300 KHz	Low Frequency (LF)or Long Wave (LW)	AM radio
300 KHz -3 MHz	Medium Frequency (MF) or Medium Wave (MW)	AM radio
3 MHz - 30 MHz	High Frequency (HF) or Short Wave (HW)	AM radio
30 MHz -300 MHz	Very High Frequency (VHF)	FM Radio-TV
300 MHz - 3 GHz	Ultra High Frequency (UHF)	TV-cellular telephony
3 GHz - 30 GHz	Super High Frequency (SHF)	Satellites
30 GHz - 300 GHz	Extra High Frequency (EHF)	Satellites-radars

Various microwave bands and their uses

Frequency	Band name	Applications
0.4 GHz - 1.5 GHz	L	Broadcasting-cellular
1.5 GHz - 5.2 GHz	S	Cellular
3.9 GHz - 6.2 GHz	C	Satellites
5.2 GHz - 10.9 GHz	X	Fixed wireless-satellite
10.9 GHz - 36 GHz	K	Fixed wireless-satellite
36 GHz - 46 GHz	Q	Fixed wireless
46 GHz -56 GHz	V	Future satellite
56 GHz -100 GHz	W	Future cellular

Wireless Media

- Transmission of waves take place in the electromagnetic (EM) spectrum
- Low frequency signals can travel for long distances through many obstacles
- high frequency signals can travel for shorter distances through few obstacles
- noise effect on the signal is inversely proportional to the power of the radio transmitter

CLASSIFICATIONS OF WIRELESS TRANSMISSION Media

Classification by Propagation

- Fixed (Directional)
- Mobile (Omnidirectional)

Classification by Method

- Radio
- Microwave
- Infrared

Applications I

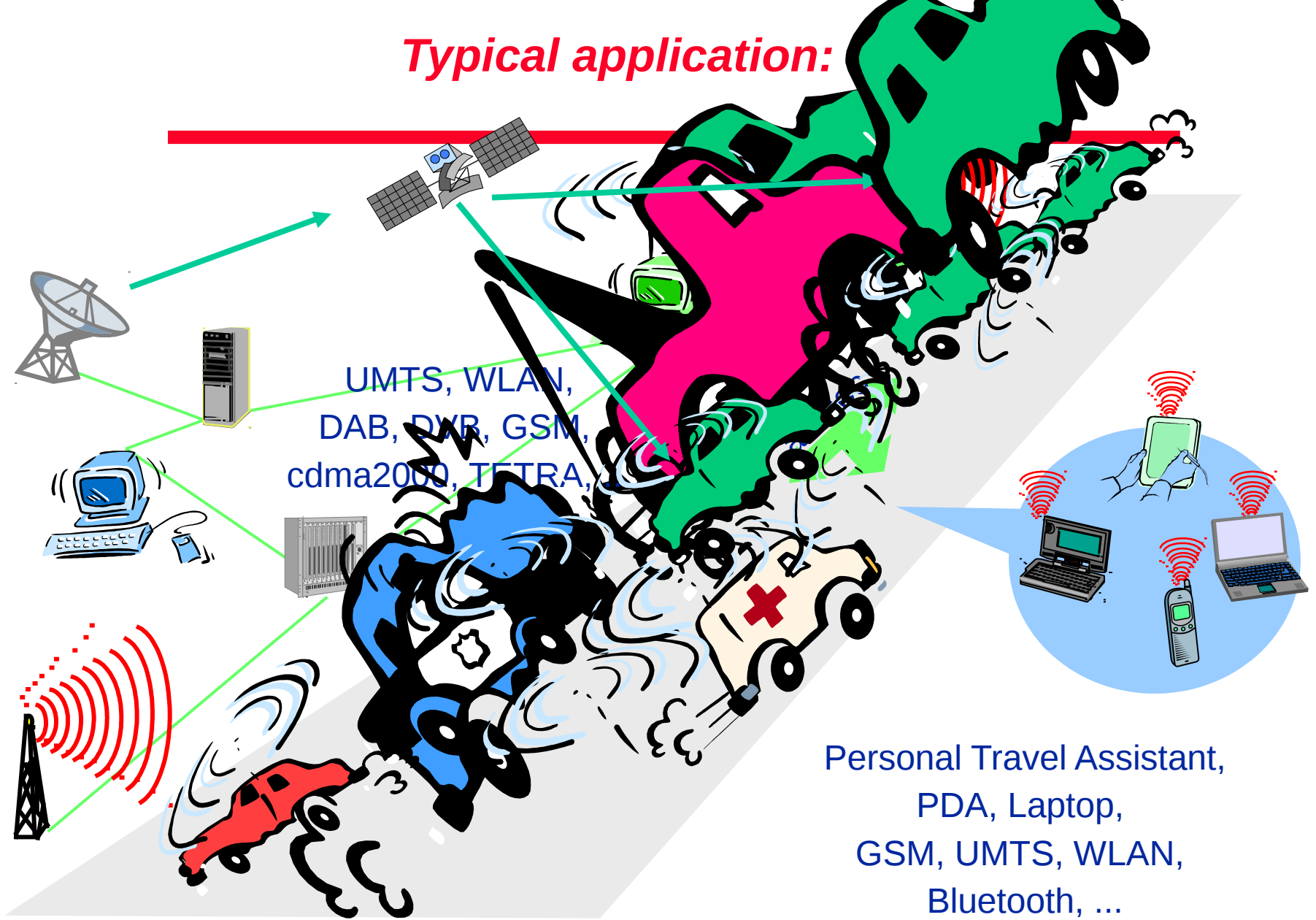
Vehicles

- ❑ transmission of news, road condition, weather, music via DAB
- ❑ personal communication using GSM
- ❑ position via GPS
- ❑ local ad-hoc network with vehicles close-by to prevent accidents, guidance system, redundancy
- ❑ vehicle data (e.g., from busses, high-speed trains) can be transmitted in advance for maintenance

Emergencies

- ❑ early transmission of patient data to the hospital, current status, first diagnosis
- ❑ replacement of a fixed infrastructure in case of earthquakes, hurricanes, fire etc.
- ❑ crisis, war, ...

Typical application:



Applications II

Travelling salesmen

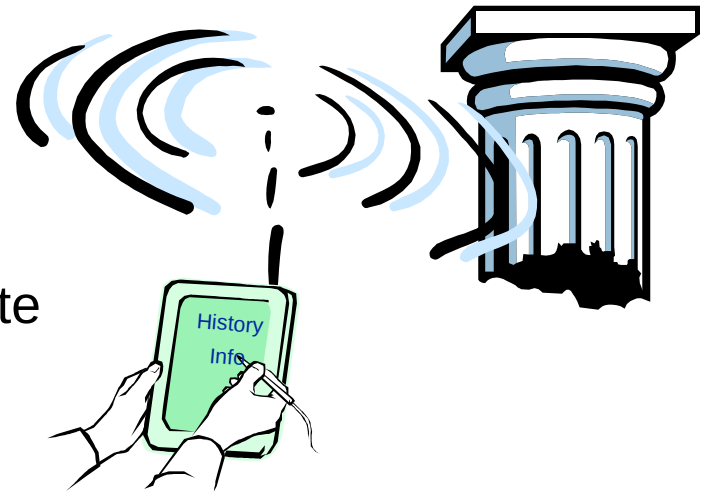
- ❑ direct access to customer files stored in a central location
- ❑ consistent databases for all agents
- ❑ mobile office

Replacement of fixed networks

- ❑ remote sensors, e.g., weather, earth activities
- ❑ flexibility for trade shows
- ❑ LANs in historic buildings

Entertainment, education, ...

- ❑ outdoor Internet access
- ❑ intelligent travel guide with up-to-date location dependent information
- ❑ ad-hoc networks for multi user games



Location dependent services

Location aware services

- ❑ what services, e.g., printer, fax, phone, server etc. exist in the local environment

Follow-on services

- ❑ automatic call-forwarding, transmission of the actual workspace to the current location

Information services

- ❑ „push“: e.g., current special offers in the supermarket
- ❑ „pull“: e.g., where is the Black Forrest Cherry Cake?

Support services

- ❑ caches, intermediate results, state information etc. „follow“ the mobile device through the fixed network

Privacy

- ❑ who should gain knowledge about the location

Mobile devices

Pager

- receive only
- tiny displays
- simple text messages



Mobile phones

- voice, data
- simple graphical displays

PDA

- graphical displays
- character recognition
- simplified WWW



Palmtop

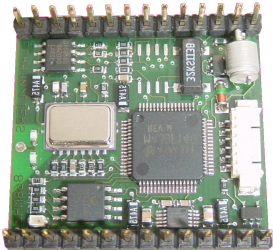
- tiny keyboard
- simple versions of standard applications

Laptop/Notebook

- fully functional
- standard applications



Sensors,
embedded
controllers



www.scatterweb.net

performance

Wireless networks in comparison to fixed networks

Higher loss-rates due to interference

- ❑ emissions of, e.g., engines, lightning

Restrictive regulations of frequencies

- ❑ frequencies have to be coordinated, useful frequencies are almost all occupied

Low transmission rates

- ❑ local some Mbit/s, regional currently, e.g., 53kbit/s with GSM/GPRS

Higher delays, higher jitter

- ❑ connection setup time with GSM in the second range, several hundred milliseconds for other wireless systems

Lower security, simpler active attacking

- ❑ radio interface accessible for everyone, base station can be simulated, thus attracting calls from mobile phones

Always shared medium

- ❑ secure access mechanisms important

Advantages of Wireless Network:

- **Mobility:** Data can be access from any location.
- **Flexible:** Radio waves can penetrate the obstacles. Sender and receiver can be placed any where.
- **Robustness :** Can survive in disaster (Earthquake, military operations)
- **Scalable:** Can be configured in variety of topologies.
- **Easy Installation**
- **Less Cost**
- **Usage of ISM band:** ISM (Industrial, Scientific and Medical) band (2.40GHz to 2.484 GHz, 5.725 GHz to 5.850 GHz) is available for use by anyone.
- **No Planning:** Only Wireless Ad hoc not required any planning.

Disadvantages of Wireless Network:

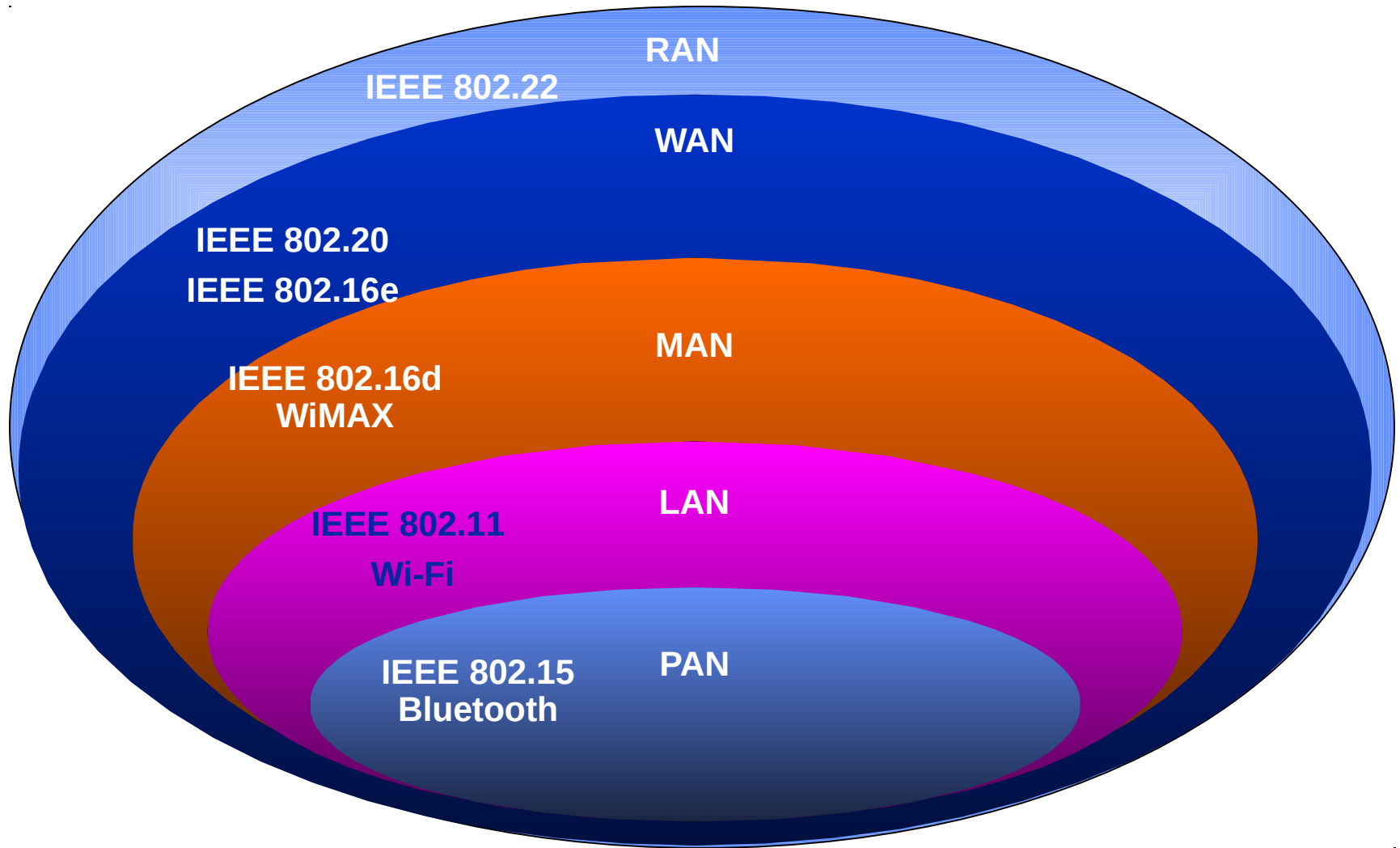
➤ **Quality of Service:**

- ✓ Lower Bandwidth
- ✓ Lower Data Transmission Rate
- ✓ High Error Rates
- ✓ Interference
- ✓ Higher Delay

Restrictions: License-free frequency bands are not same worldwide.

➤ **Safety and Security:** Interference from other devices (e.g. Hospital. Eavesdropping is possible).

IEEE Wireless Standards



Challenges

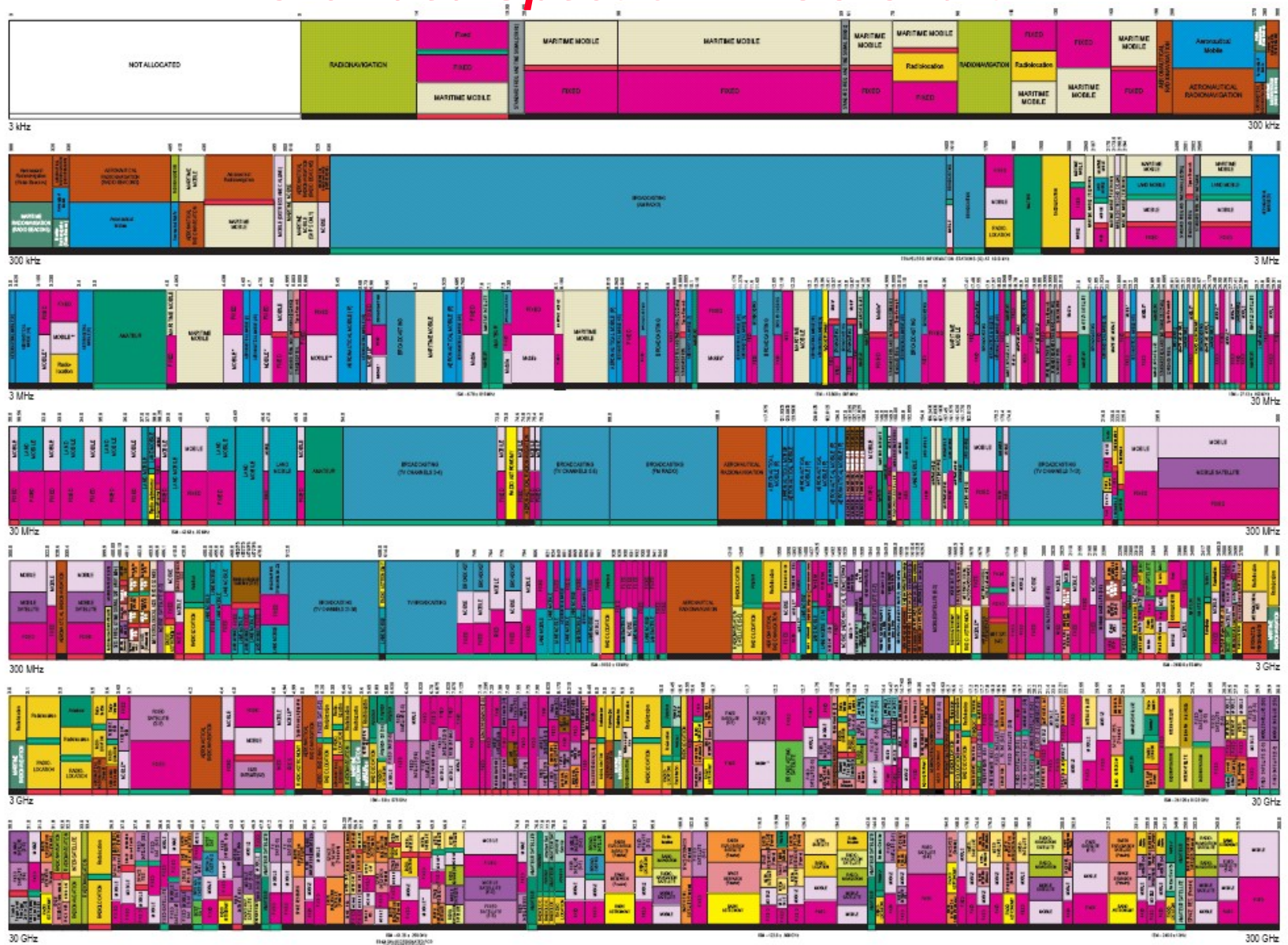
Challenge 1: Unreliable and Unpredictable Wireless Coverage

- Wireless channel “feels” very different from a wired channel.
 - Wireless links are not reliable: they may vary over time and space
 - Noise adds on to the signal
 - Signal strength falls off rapidly with distance
 - Signal strength may weaken due to obstacles
 - Medium “air” shared among many users
- Results:
 - Capacity is shared with others
 - Variable capacity
 - Unreliable channel: errors, outages
 - Variable delays

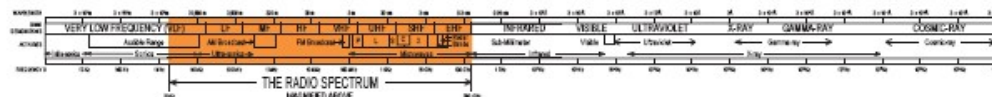
Challenge 2: Spectrum use

- Wireless spectrum is scarce resource
- Expensive
- Efficient regulation required(FCC, ETSI)
- Licensed spectrum
- System capacity over given band

Crowded Spectrum: FCC Chart



* EXCEPT WHERE SHOWN
* EXCEPT WHERE SHOWN



PLEASE NOTE: THE FREQUENCY ALLOCATIONS IN THIS CHART
DO NOT REPRESENT THE FREQUENCY ALLOCATIONS IN ALL COUNTRIES
OF THE WORLD.

Challenge 3: Power Management

- Hosts in WN are powered by batteries
 - Finite storage capacity of battery requires recharging
 - System with increased time of operation between successive recharges are needed
 - Batteries should not weigh a lot
 - P is proportional to CV^2F
 - VLSI to be used to C , V or F to increase efficiency of battery
 - Wireless systems should be built in such a way that most of processing is carried out at fixed part of network
 - Efficient software can also increase battery life

Challenge 4: Security

- Interception of signal is easy in wireless transmission compared to wireline
- High level of security is required for banking and e-commerce application over WN

Challenge 5: Location/Routing(Mobility)

- Topology is not static
- Mobility causes poor-quality wireless links
- Efficient techniques required for
 - Locating MT
 - Support ongoing call during handoff
- Routing becomes difficult in adhoc networks

Challenge 6: Interfacing with wired networks

- This is required for mobile users to use all different applications like voice call to PSTN, file transfer, e-mail, Internet access etc.

Challenge 7: Health Concerns

- ?????